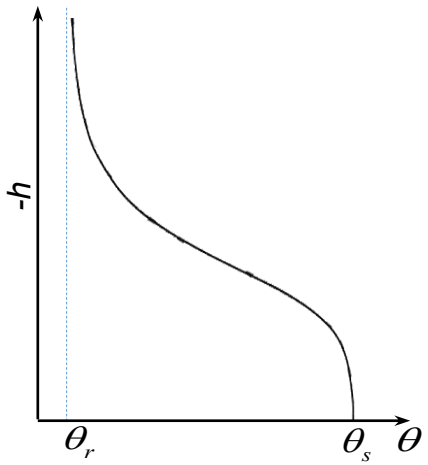
HWRS 561b: Physical Hydrogeology II

PFAS (part 2)

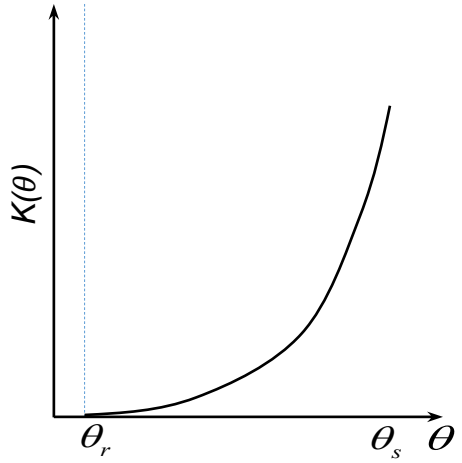
Modeling PFAS Transport in the Vadose Zone

Variably saturated flow in the vadose zone

$$\frac{\partial \theta}{\partial t} - \nabla \cdot [\boldsymbol{K}(\theta)\nabla(h - z)] = 0$$



SWC



Unsat hydraulic conductivity

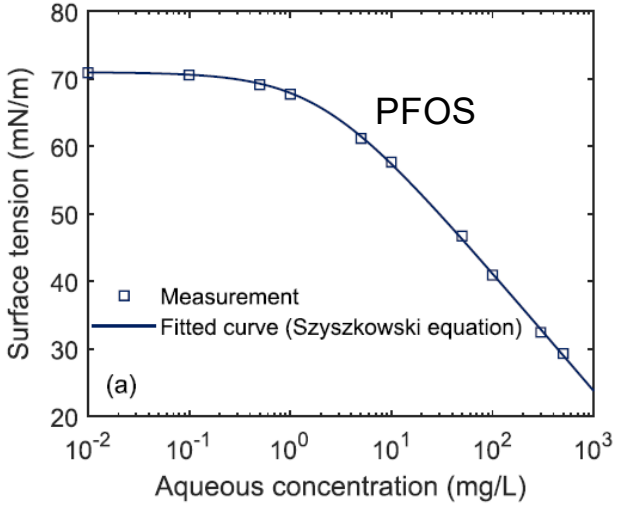
PFAS dissolving in water changes surface tension

$$\sigma = \sigma_0[1 - b\ln(1 + \frac{C}{a})]$$

PFAS concentration

Surface tension of water w/o PFAS

Fitting parameters



Question: Will the SWC and $K - \theta$ change in the presence of PFAS? If so, how?

Modeling PFAS Transport in the Vadose Zone

Let's employ the capillary tube model to think about the problem (Leverett-J scaling)

w/o PFAS

$$-h_0 = \frac{2\sigma_0 \cos\gamma_0}{\rho g r}$$

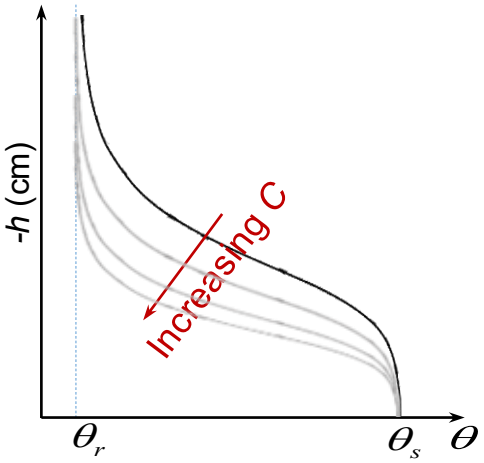
w/ PFAS

$$-h = \frac{2\sigma \cos\gamma}{\rho g r}$$

$$\Rightarrow h = \frac{\sigma \cos\gamma}{\sigma_0 \cos\gamma_0} h_0$$

$$\Rightarrow \theta = \begin{cases} \theta_r + \dfrac{\theta_s - \theta_r}{\left[1 + \left(\dfrac{\sigma_0 \cos\gamma_0}{\sigma \cos\gamma}\alpha|h|\right)^n\right]^m}, h < 0 \\ \theta_s, h \geq 0 \end{cases}$$

Scaling factor



Insights:

(1) We can think of this as PFAS modifies the α in the V-G model.

$$\alpha_{\text{modified}} = \frac{\sigma_0 \cos\gamma_0}{\sigma \cos\gamma}\alpha$$

(2) Now, the variably saturated flow depends on PFAS transport.

Modeling PFAS Transport in the Vadose Zone

PFAS transport in variably unsaturated flow (neglecting the interactions among PFAS)

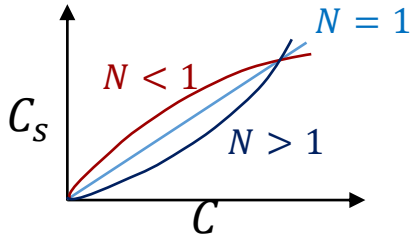
Advection Dispersion

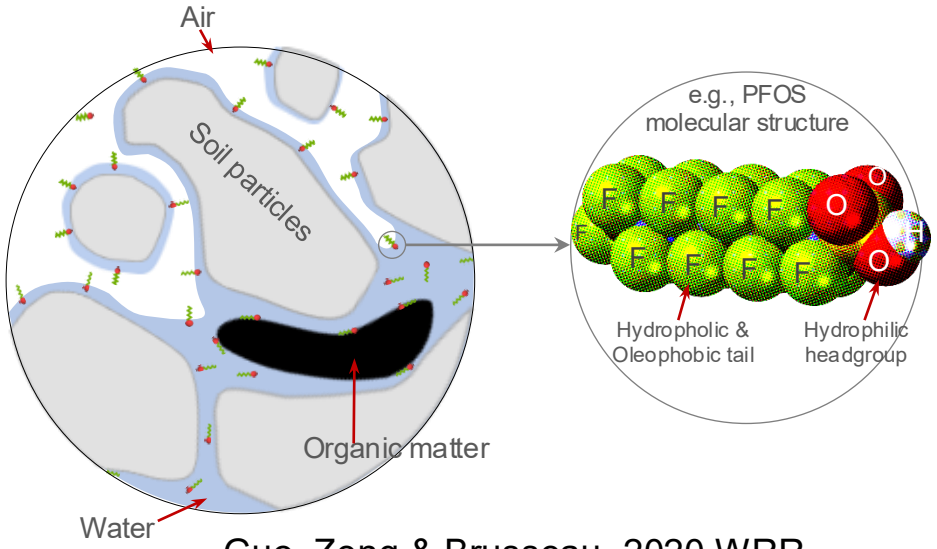
$$\frac{\partial(\theta C)}{\partial t} + \rho_b \frac{\partial C_s}{\partial t} + \frac{\partial C_{aw}}{\partial t} + \nabla \cdot (\theta \boldsymbol{v} C) - \nabla \cdot (\theta \boldsymbol{D} \nabla C) = 0$$

Storage in the aqueous phase

Storage in the solid-phase adsorption

Storage in the air-water interfacial adsorption





Guo, Zeng & Brusseau. 2020 WRR

Solid-phase adsorption $C_s = K_f C^N$

Air-water interfacial area

Air-water interfacial adsorption $C_{aw} = \Gamma A_{aw} = K_{aw} C A_{aw}$ $\sigma = \sigma(C)$ Szyszkowski equation

$$K_{aw} = \frac{\Gamma}{C} = -\frac{1}{RTC}\left(\frac{\partial \sigma}{\partial \ln C}\right)_T$$

$$\Rightarrow K_{aw} = \frac{1}{RT}\frac{\sigma_0 b}{a + C}$$

Modeling PFAS Transport in the Vadose Zone

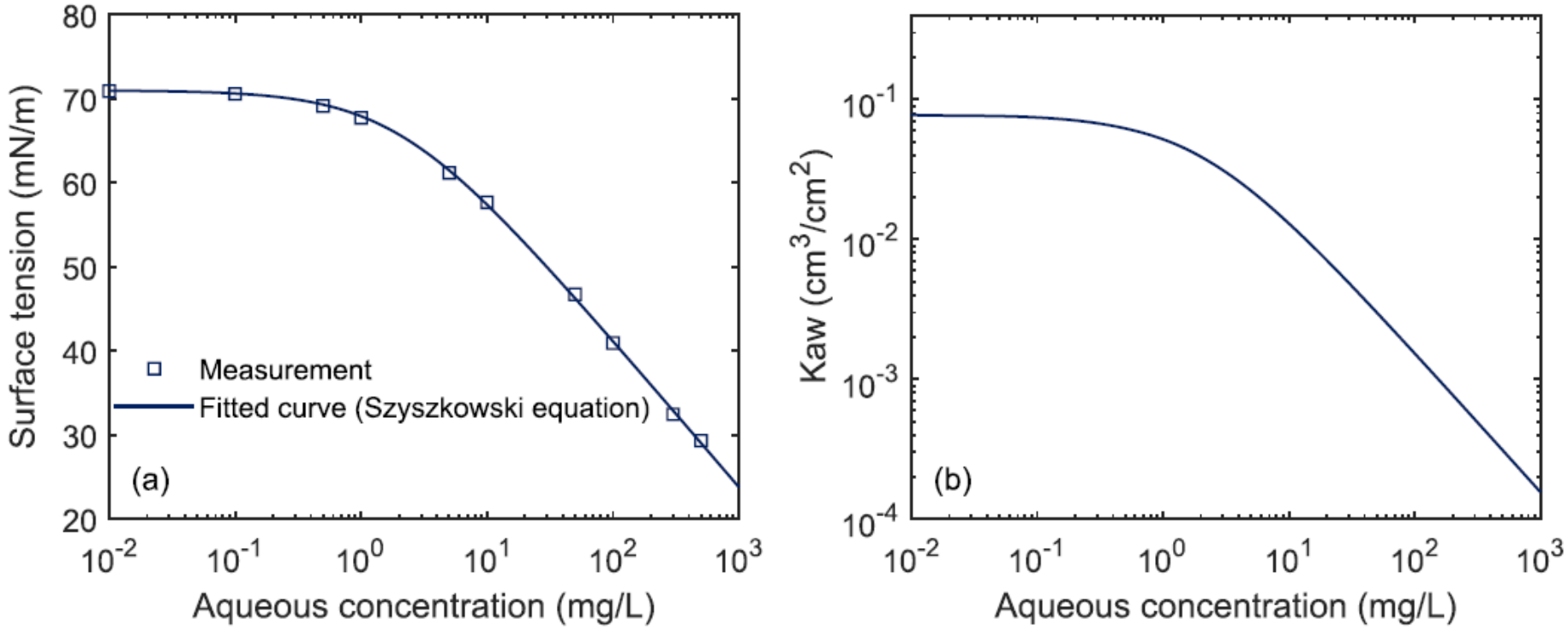


Figure 3. (a) Measured surface tension as a function of the aqueous concentration of PFOS. The solid line is a fitted curve using the Szyszkowski equation. (b) The computed air-water interfacial adsorption coefficient (K_{aw}) as a function of the aqueous concentration of PFOS.

Guo, Zeng & Brusseau. 2020 WRR

Simplified Models

- Steady-state unsaturated flow:

$$q = K$$

- PFAS transport:

$$\frac{\partial(\theta C)}{\partial t} + \rho_b \frac{\partial C_s}{\partial t} + \frac{\partial C_{aw}}{\partial t} + \nabla \cdot (\theta \boldsymbol{v} C) - \nabla \cdot (\theta \boldsymbol{D} \nabla C) = 0$$

Advection Dispersion

All the equations are linearized and can be solved analytically (Guo et al., 2022). The solutions are in the forms of integrals, which can be implemented in Excel spreadsheet (See additional details in the PFAS-LEACH Excel User Guide (Ma et al. 2025).

Linear and rate-limited solid-phase adsorption

$$C_{s,1} = F_s K_d C$$

Instantaneous

$$\frac{dC_{s,2}}{dt} = \alpha_s[(1 - F_s)K_d C - C_{s,2}]$$

Kinetic

Linear and equilibrium air-water interfacial adsorption

$$C_{aw} = A_{aw} K_{aw} C$$

Instantaneous

Kaw can be assumed as a constant at lower concentrations, which is often the case at PFAS-contaminated sites

